COMPLAINT COMPASS · WHEN THEY ASK "HOW WILL YOU SCALE THIS?"

Scalability script (about 50 seconds)

Bold words carry the emphasis. Throughput and server figures are projected from latencies measured on the live run.

> "We scale on three axes: **volume, customers, and spikes**. These are projections from latencies we measured on our live run.
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> **Volume.** In production the system is two tiers joined by a queue. The **classification tier is stateless**, so we autoscale it on queue lag. One worker handles about **15 complaints a second**, roughly **1.3 million a day**. All of Indian quick commerce averages about **2 a second**. So one business at up to **a million complaints a month** runs on **one or two servers**, for about **$43 a month** in model cost.
>
> **Customers.** The issue registry is partitioned by **business and area**, with one writer per partition. Each partition handles about **3.5 complaints a second**, so a grocery business's 7 areas handle about **24 a second**. At **10 million a month**, about 10 businesses, that's about **70 partitions** and **2 to 4 workers**, for about **$433 a month**. At **100 million**, about **700 partitions** and **13 to 16 workers**, for about **$4,330 a month**.
>
> **Spikes.** We plan for **5× the average**. Intake never stops, because complaints queue. If a partition lags more than **30 seconds**, we split it by region.
>
> We prove all this with a **load test: 20 complaints a second for an hour**, with zero duplicates and zero lost."

The numbers in front of you

Volume a month	Average / peak per second	Classifier workers	Partitions	Model cost a month
1M (one business)	0.4 / 2	2	7	$43
10M (~10 businesses)	3.9 / 19	2–4	~70	$433
100M (~100 businesses)	39 / 193	13–16	~700	$4,330

Where they come from, if a judge asks

- **15 a second per worker:** 8 Jev calls at a time, each taking 0.53 s (measured).
- **3.5 a second per partition:** the measured mix of match (0.02 s), Jev check (0.54 s, 35% of complaints) and LLM (1.9 s, about 4%), handled one at a time.
- **About 2 a second for the whole market:** 7.8 million orders a day (RedSeer), times our assumption that 2% lead to a complaint.
- **Model cost:** $4.33 per 100,000 complaints (measured) times the volume. Hosting is not included.

The 5× peak is an assumption, and none of this is load-tested yet. If asked: "Projected, and the load test is our proof point."